Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/29/2026
PI Name:	Strock, Nicole Ph.D.		
Project Title:	Functional Task Test Data Analysis With Respect to Muscle Strength and Aerobic Capacity Adaptations Following Spaceflight and Bed Rest		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Muscle and Aerobic**:Risk of Mission Performance Decrements and Adverse Health Outcomes Resulting from Reduced Cardiorespiratory and Musculoskeletal Fitness		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	KBR		
PI Address 1:	JSC B21 / 1216F		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	11/01/2023	**End Date:**	09/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Additional personnel were involved in this investigation as of 2025: Jonathan Hummel, John Hardy, Ethan Tinoco, Brian Prejean, Millennia Young (augmentation), Thomas Oswald (augmentation), Lori Chappell (augmentation).		
COI Name (Institution):	Varanoske, Alyssa (KBR/NASA) Frisco, Dillon (JES Tech/NASA) Hummel, Jonathan (University of Houston/NASA)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Previous investigations, as well as ground-based spaceflight analogs, have demonstrated that exposure to weightlessness results in decreased work and exercise capacity, including reduced muscle strength and endurance, aerobic capacity, and ventilatory threshold. However, analyses testing the relationship between these physiological measures of decreased performance to surface mission tasks, similar to what astronauts will be expected to perform in the first days and weeks of activities on the Moon and Mars, have not been completed. Data from the NASA Functional Task Test and Field Test studies, combined with astronaut medical testing (MEDB) aerobic capacity and muscle strength measures, will be leveraged to address this knowledge gap. This project builds upon previous work (Exercise Requirements for Gateway and Orion: Human Physiology, Performance, Protection, and Operations (H-3PO) Laboratory Review of Muscle and Aerobic Outcomes in Relation to Exercise Prescription Design) that identified “responders” and “non-responders” to inflight exercise countermeasures in the astronaut population. An important aspect of this analysis will include consideration of these two groups on functional performance outcomes measured after spaceflight and bed rest. The team will: 1) Collate functional task results from the Functional Task Test (spaceflight and bed rest) and Field Test studies, and collate muscle strength, endurance, aerobic capacity, and ventilatory threshold test results for subjects from either study data and/or MedB assessments. 2) Use statistical modeling to test for relationships between physiological measures and functional task outcomes, including immediate post-spaceflight and post-bed rest measures, as well as during recovery. 3) Provide a report to the NASA Human Research Program (HRP) detailing the methods, results, and interpretation of these analyses. The interpretation should include how these results impact the muscle and aerobic standards for partial gravity, and requirements for exercise countermeasures across relevant design reference missions. The report also will include limitations of these findings and how the results impact or inform future work. As of 2025, an augmentation to this project was included. Constraints imposed by Artemis IV mission timelines, restrictions on exercise on Orion during docked operations, and delays in the availability of the Gateway exercise device likely will result in days or weeks without significant musculoskeletal and cardiovascular countermeasures before the crew descend to the lunar surface. Though many spaceflight analog studies (e.g., bed rest, dry immersion) have documented a reduction in physiological measures of cardiopulmonary and musculoskeletal capacity (e.g. peak oxygen consumption, leg extensor strength), little data exist for astronauts who perform little to no exercise during spaceflight. Further, there have been few attempts to extrapolate these physiologic measures to task performance metrics that are operationally relevant and meaningful to mission planners. Recently, the Moon to Mars Program requested that the SK/Exercise Physiology and Countermeasures (EPC) Laboratory work with the SK/Biostatistics Laboratory in the development of a statistical method to predict physiological and performance using existing data from spaceflight and spaceflight analogs where available, particularly those including measures of aerobic and muscular fitness as well as functional EVA-relevant task performance outcomes. Specific Aims To develop a statistical analysis model to predict EVA-relevant functional performance across 0-30 days of spaceflight deconditioning without exercise by using results from spaceflight and spaceflight analogs in peer-reviewed publications and available individual data from NASA studies (LSAH and PI science). This will be accomplished by: 1. Aggregating physiological performance data (e.g., VO2pk, isokinetic leg extensor strength) from multiple spaceflight analog studies (no less than 15 studies) ranging from 7-30+ days in duration in which participants performed no exercise 2. Developing a statistical analysis model for each outcome variable (e.g., VO2pk, leg extensor strength) relating number of days without exercise to physiological performance results 3. From these outcomes, predict functional performance outcomes (e.g., capsule egress, incapacitated crewmember recovery) using available data relating strength and aerobic fitness metrics to measures of functional task performance
Rationale for HRP Directed Research:	This task is directed because it contains highly constrained research.
Research Impact/Earth Benefits:	
Task Progress:	In this reporting period, data analysis was completed for the original project and augmentation. Findings were described in two final reports.
Bibliography Type:	Description: (Last Updated: 04/09/2026)
Abstracts for Journals and Proceedings	Strock NC, Frisco DJ, Varanoske AN, Allen JR, Hummel JW, Prejean BJ. "Neurovestibular functional task test data analysis with respect to muscle strength and aerobic capacity adaptations following spaceflight." 2025 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025